

MAT 319 HW02

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Due 9/11

1 Textbook Problems

Problem 7.1. Write out the first five terms of the following sequences.

(a) $s_n = \frac{1}{3n+1}$

(b) $b_n = \frac{3n+1}{4n-1}$

(c) $c_n = \frac{n}{3^n}$

(d) $\sin(\frac{n\pi}{4})$

Solution. Part (a): $\frac{1}{4}, \frac{1}{7}, \frac{1}{10}, \frac{1}{13}, \frac{1}{16}$

Part (b): $\frac{4}{3}, \frac{7}{7}, \frac{10}{11}, \frac{13}{15}, \frac{16}{19}$

Part (c): $\frac{1}{3}, \frac{2}{9}, \frac{3}{27}, \frac{4}{81}, \frac{5}{243}$

Part (d): $\frac{\sqrt{2}}{2}, 1, \frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}$

□

Problem 7.3. For each sequence below, determine whether it converges and, if it converges, give its limit. No proofs are required.

(a) $a_n = \frac{n}{n+1}$

(c) $c_n = 2^{-n}$

(e) $x_n = 73 + (-1)^n$

(g) $y_n = n!$

(i) $\frac{(-1)^n}{n}$

(k) $\frac{9n^2-18}{6n+18}$

(m) $\sin(n\pi)$

(o) $\frac{\sin n}{n}$

(q) $\frac{3^n}{n!}$

(s) $\frac{4n^2+3}{3n^2-2}$

(b) $b_n = \frac{n^2+3}{n^2-3}$

(d) $t_n = 1 + \frac{2}{n}$

(f) $s_n = 2^{1/n}$

(h) $d_n = (-1)^n n$

(j) $\frac{7n^3+8n}{2n^3-3}$

(l) $\sin(\frac{n\pi}{2})$

(n) $\sin(\frac{2n\pi}{3})$

(p) $\frac{2^{n+1}+5}{2^n-7}$

(r) $(1 + \frac{1}{n})^2$

(t) $\frac{6n+4}{9n^2+7}$

Solution.

(a) Converges to 1.

(c) Converges to 0.

(e) Does not converge.

(g) Does not converge (diverges to $+\infty$).

(i) Converges to 0.

(k) Does not converge (diverges to $+\infty$).

(m) Converges to 0.

(o) Converges to 0.

(q) Converges to 0.

(s) Converges to $\frac{4}{3}$.

(b) Converges to 1.

(d) Converges to 1.

(f) Converges to 1.

(h) Does not converge.

(j) Converges to $\frac{7}{2}$.

(l) Does not converge.

(n) Does not converge.

(p) Converges to 2.

(r) Converges to 1.

(t) Converges to 0.

□

Problem 8.1. Prove the following:

(a) $\lim_{n \rightarrow \infty} \frac{(-1)^n}{n} = 0$

(b) $\lim_{n \rightarrow \infty} \frac{1}{n^{1/3}} = 0$

(c) $\lim_{n \rightarrow \infty} \frac{2n-1}{3n+2} = \frac{2}{3}$

Solution. For part (a), we want to show that, for every $\varepsilon > 0$,

$$\left| \frac{(-1)^n}{n} - 0 \right| < \varepsilon$$

is eventually true. We show that as long as $n > \frac{1}{\varepsilon}$, the inequality holds; indeed,

$$\left| \frac{(-1)^n}{n} - 0 \right| = \left| \frac{(-1)^n}{n} \right| = \frac{1}{n} < \frac{1}{1/\varepsilon} = \varepsilon,$$

so the inequality is eventually true and the limit of the sequence is 0.

For part (b), we want to show that, for every $\varepsilon > 0$,

$$\left| \frac{1}{n^{1/3}} - 0 \right| < \varepsilon$$

is eventually true. We show that as long as $n > \frac{1}{\varepsilon^3}$, the inequality holds; indeed,

$$\left| \frac{1}{n^{1/3}} - 0 \right| = \left| \frac{1}{n^{1/3}} \right| = \frac{1}{n^{1/3}} < \frac{1}{(1/\varepsilon^3)^{1/3}} = \frac{1}{1/\varepsilon} = \varepsilon,$$

so the inequality is eventually true and the limit of the sequence is 0.

For part (c), we want to show that, for every $\varepsilon > 0$,

$$\left| \frac{2n-1}{3n+2} - \frac{2}{3} \right| < \varepsilon$$

is eventually true. We show that as long as $n > \frac{7}{9\varepsilon} - \frac{2}{3}$, the inequality holds; indeed,

$$\begin{aligned} \left| \frac{2n-1}{3n+2} - \frac{2}{3} \right| &= \left| \frac{3(2n-1) - 2(3n+2)}{3(3n+2)} \right| \\ &= \left| \frac{-7}{9n+6} \right| = \frac{7}{9n+6} \\ &< \frac{7}{9\left(\frac{7}{9\varepsilon} - \frac{2}{3}\right) + 6} \\ &= \frac{7}{7/\varepsilon - 6 + 6} = \varepsilon, \end{aligned}$$

so the inequality is eventually true and the limit of the sequence is $\frac{2}{3}$. □

Problem 8.2. Determine, with proof, the limits of the following sequences.

(a) $a_n = \frac{n}{n^2+1}$

(b) $b_n = \frac{7n-19}{3n+7}$

(c) $c_n = \frac{4n+3}{7n-5}$

Solution. The limit of a_n is $\boxed{0}$. To prove this, we want to show that, for all $\varepsilon > 0$,

$$|a_n - 0| < \varepsilon$$

is eventually true. I claim the inequality holds whenever $n > \frac{1}{\varepsilon}$. Indeed, the condition implies

$$\left| \frac{n}{n^2+1} - 0 \right| = \frac{n}{n^2+1} < \frac{n}{n^2} = \frac{1}{n} < \frac{1}{1/\varepsilon} = \varepsilon,$$

so the inequality is eventually true and thus the limit is 0.

The limit of b_n is $\boxed{\frac{7}{3}}$. To prove this, we want to show that, for all $\varepsilon > 0$,

$$\left| b_n - \frac{7}{3} \right| < \varepsilon$$

is eventually true. I claim the inequality holds whenever $n > \frac{106}{9\varepsilon} - \frac{7}{3}$. Indeed, the condition implies

$$\begin{aligned} \left| \frac{7n-19}{3n+7} - \frac{7}{3} \right| &= \left| \frac{3(7n-19) - 7(3n+7)}{3(3n+7)} \right| \\ &= \left| \frac{-106}{9n+21} \right| = \frac{106}{9n+21} \\ &< \frac{106}{9\left(\frac{106}{9\varepsilon} - \frac{7}{3}\right) + 21} \\ &= \frac{106}{106/\varepsilon - 21 + 21} = \varepsilon, \end{aligned}$$

so the inequality is eventually true and thus the limit is $\frac{7}{3}$.

The limit of c_n is $\boxed{\frac{4}{7}}$. To prove this, we want to show that, for all $\varepsilon > 0$,

$$\left| c_n - \frac{4}{7} \right| < \varepsilon$$

is eventually true. I claim the inequality holds whenever $n > \frac{43}{49\varepsilon} + \frac{5}{7}$. Indeed, the condition implies

$$\begin{aligned} \left| \frac{4n+3}{7n-5} - \frac{4}{7} \right| &= \left| \frac{7(4n+3) - 4(7n-5)}{7(7n-5)} \right| \\ &= \left| \frac{43}{49n-35} \right| = \frac{43}{49n-35} \\ &< \frac{43}{49\left(\frac{43}{49\varepsilon} + \frac{5}{7}\right) - 35} \\ &= \frac{43}{43/\varepsilon + 35 - 35} = \varepsilon, \end{aligned}$$

so the inequality is eventually true and thus the limit is $\frac{4}{7}$. □

Problem 9.13. Show that

$$\lim_{n \rightarrow \infty} a^n = \begin{cases} 0 & \text{if } |a| < 1 \\ 1 & \text{if } a = 1 \\ +\infty & \text{if } a > 1 \\ \text{does not exist} & \text{if } a \leq -1. \end{cases}$$

Solution. Clearly if $a = 1$ then $a^n = 1^n = 1$ for all n , so that case is obvious.

If $a > 1$, then we want to show that $\lim_{n \rightarrow \infty} a^n = +\infty$, which means that, for each $M \in \mathbb{R}$, a^n is eventually always greater than M . To show this, write $a = 1 + x$ for a positive real x . Then,

$$a^n = (1 + x)^n = 1 + nx + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \cdots + x^n > 1 + nx$$

by the binomial theorem. This means that, if $n > \frac{M-1}{x}$,

$$a^n > 1 + nx = M,$$

so a^n is eventually greater than M , meaning $(a, a^2, a^3, \dots)$ diverges to $+\infty$.

Now, we treat the case of $|a| < 1$ by splitting it into the three cases $a = 0$, $0 < a < 1$, and $-1 < a < 0$.

- If $a = 0$, then $a^n = 0$ for all n , so the limit is 0 (because $\varepsilon > 0$ means that $\varepsilon > 0 = |a^n - 0|$).
- If $0 < a < 1$, suppose for the sake of contradiction that $\lim_{n \rightarrow \infty} a^n$ is nonzero, say equal to some positive real c (c must be positive because $a^n > 0$ for all n). Then, for all $\varepsilon > 0$, the inequality $|a^n - c| < \varepsilon$ is eventually true, say for all $n > N$. This means that for $n > N$, $a^n < c + \varepsilon$ because

$$c + \varepsilon > |c| + |a^n - c| \geq |a^n| = a^n,$$

so, for all such n ,

$$\left| \frac{1}{a^n} - \frac{1}{c} \right| = \left| \frac{c - a^n}{ca^n} \right| < \frac{\varepsilon}{c(c + \varepsilon)} < \frac{\varepsilon}{c^2}$$

where the first inequality comes from $|c - a^n| < \varepsilon$ and $\frac{1}{c + \varepsilon} < \frac{1}{a^n}$. I claim that this means

$$\lim_{n \rightarrow \infty} \frac{1}{a^n} = \frac{1}{c}.$$

To show this, we want to prove that for all $\varepsilon' > 0$,

$$\left| \frac{1}{a^n} - c \right| < \varepsilon'.$$

We showed above that, for any $\varepsilon > 0$,

$$\left| \frac{1}{a^n} - \frac{1}{c} \right| > \frac{\varepsilon}{c^2},$$

so picking $\varepsilon = c^2\varepsilon'$ shows that there is an N' such that $n > N'$ implies

$$\left| \frac{1}{a^n} - c \right| < \varepsilon',$$

which is enough to show that the limit of $\frac{1}{a^n}$ as n goes to infinity is $\frac{1}{c}$. This, finally, is our contradiction: we have shown that the limit of $(\frac{1}{a})^n$ is $\frac{1}{c}$, which means the sequence given by $(\frac{1}{a})^n$ should converge; however, we showed earlier that, since $\frac{1}{a} > 1$, this sequence diverges to $+\infty$! Thus, $\lim_{n \rightarrow \infty} a^n = 0$, as desired.

- For the last case, $-1 < a < 0$, we want to show that, for each $\varepsilon > 0$,

$$|a^n - 0| = |a^n| = |a|^n < \varepsilon$$

is eventually true. This is equivalent to showing that

$$||a|^n - 0| < \varepsilon$$

is eventually true — that is, it suffices to show that the limit as n goes to infinity of $|a|^n$ is zero. This follows directly from the case where $0 < a < 1$ because $|a|$ is between 0 and 1, so we are done!

Finally, we need to show that $a < -1$ implies that $\lim_{n \rightarrow \infty} a^n$ doesn't exist. Suppose for the sake of contradiction that the limit does exist. Then, the limit is either a real number or $\pm\infty$. In the former case, let the limit be L . This means that for all $\varepsilon > 0$,

$$|a^n - L| < \varepsilon$$

is eventually true, so it must be true for $\varepsilon = \frac{1}{2}$. We find by the triangle inequality that

$$1 = 2\varepsilon > |a^n - L| + |a^{n+1} - L| = |a^n - L| + |L - a^{n+1}| \geq |a^n - a^{n+1}| = |a|^n + |a|^{n+1}$$

with the last equality following from $a < 0$, since a^n and a^{n+1} have different signs. Then, since $|a| > 1$, $|a|^m > 1$ for all m , which means

$$1 = 2\varepsilon > |a|^n + |a|^{n+1} > 1 + 1 = 2,$$

which is a contradiction.

Therefore, if the limit exists, it must be one of $+\infty$ and $-\infty$. This means either for all real M , $a^n > M$ eventually, or for all real M , $a^n < M$ eventually. Taking $M = 0$ shows that statement is false, since $a^n < 0$ if n is odd and $a^n > 0$ if n is even, so $a^n < 0$ and $a^n > 0$ are both frequently true and thus both are also frequently false (meaning neither is eventually true). This shows that the limit doesn't exist if $a < -1$. $\square$

Problem 9.18.

- (a) Verify $1 + a + a^2 + \cdots + a^n = \frac{a^{n+1}-1}{a-1}$ for all $a \neq 1$.
- (b) Find $\lim_{n \rightarrow \infty} (1 + a + a^2 + \cdots + a^n)$ for $|a| < 1$.
- (c) Calculate $\lim_{n \rightarrow \infty} (1 + \frac{1}{3} + \frac{1}{9} + \cdots + \frac{1}{3^n})$.
- (d) What is $\lim_{n \rightarrow \infty} (1 + a + a^2 + \cdots + a^n)$ for $a \geq 1$?

Solution. For part (a), we can see that

$$\begin{aligned} (a-1)(1 + a + a^2 + \cdots + a^n) &= a(1 + a + a^2 + \cdots + a^n) - (1 + a + a^2 + \cdots + a^n) \\ &= (a + a^2 + a^3 + \cdots + a^{n+1}) - (1 + a + a^2 + \cdots + a^n) \end{aligned}$$

and then $a, a^2, \dots, a^n$ all cancel out, leaving behind just $a^{n+1} - 1$; dividing by $a - 1$ shows the desired equality. (Alternatively, induction also works nicely.)

For part (b), I claim that

$$\lim_{n \rightarrow \infty} (1 + a + a^2 + \cdots + a^n) = \boxed{\frac{1}{1-a}}$$

for $|a| < 1$. To see this, we simplify the expression in the limit via part (a), at which point it suffices by the definition of limit to prove that, for all $\varepsilon > 0$,

$$\left| \frac{a^{n+1}-1}{a-1} - \frac{1}{1-a} \right| < \varepsilon$$

is eventually true. This is fairly easy to prove: it ends up being equivalent to show that

$$\varepsilon > \left| \frac{a^{n+1}-1}{a-1} - \frac{1}{1-a} \right| = \left| \frac{a^{n+1}}{a-1} \right| \iff |a-1|\varepsilon > |a^{n+1}| = a^{n+1}$$

is eventually true. We know from Problem 9.13 that the limit of a^n as n goes to infinity is 0 when $|a| < 1$, so, for each $\varepsilon' > 0$, there exists an N such that $n > N$ implies $|a^n - 0| = a^n < \varepsilon'$. Taking $\varepsilon' = |a-1|\varepsilon$ shows that

$$|a-1|\varepsilon > a^{n+1}$$

is true for $n > N$ (noting that $n+1 > n > N$), i.e. the desired inequality is eventually true. Therefore, the limit is indeed $\frac{1}{1-a}$.

Part (c) is just part (b) in the case where $a = \frac{1}{3}$ (note that $|\frac{1}{3}| < 1$, which is the necessary condition for the expression in part (b) to work), so the desired limit is just

$$\frac{1}{1-\frac{1}{3}} = \boxed{\frac{3}{2}}.$$

Finally, for part (d), the answer is $+\infty$. We see that $a \geq 1$ implies $a^n \geq 1$ for all a — this is easy to show by e.g. induction, since $a^n \geq 1$ implies $a^{n+1} = a^n \cdot a \geq 1$. Therefore,

$$1 + a + a^2 + \cdots + a^n \geq n$$

for all n . This means that, for each real M , if N is the smallest natural number greater than M , then $n > N$ implies

$$1 + a + a^2 + \cdots + a^n \geq n > N > M,$$

which is exactly what it means for the limit to be $+\infty$. $\square$

2 The Other Problems

Problem 1. Say if the following statements are true or false. Carefully justify your answer.

- (1) A statement $P(n)$, $n \in \mathbb{N}$, is true frequently if and only if it is true for infinitely many values of n .
- (2) $\sqrt{n} \in \mathbb{N}$ frequently.
- (3) $\sqrt{n} \in \mathbb{Q}$ frequently.

Solution. (1) is true. We prove the equivalent statement that $P(n)$ is not true frequently (i.e. is eventually false) if and only if $P(n)$ is true for finitely many n .

First, if $P(n)$ is eventually false, then by definition there is an N such that $n > N$ implies $P(n)$ is false. This means that if $P(n)$ is true then $n \leq N$, but, since there are only finitely many natural numbers less than N , $P(n)$ can only be true for finitely many n .

Conversely, if $P(n)$ is true for finitely many n , then the set of n for which $P(n)$ is true has some largest value N . In particular, $n > N$ implies $P(n)$ is false, i.e. $P(n)$ is eventually false as desired.

(2) and (3) are both true. We just prove (2), since $\sqrt{n} \in \mathbb{N}$ frequently implies $\sqrt{n} \in \mathbb{Q}$ frequently because $\mathbb{N} \subseteq \mathbb{Q}$. To prove (2), part (1) says that we just need to find an infinite set of numbers whose square roots are positive integers; the obvious set $\{1^2, 2^2, 3^2, 4^2, \dots\}$ does the job. $\square$

Problem 2. For $a_n = n(n+1)$, say if the following statements are true or false.

- (1) a_n is eventually even,
- (2) a_n is eventually odd,
- (3) a_n is frequently even,
- (4) a_n is frequently odd.

Solution. (1) and (3) are true, (2) and (4) are false.

We begin by observing that n and $n+1$ are consecutive integers, so one of them must be odd and the other must be even regardless of the value of n . This means that $n(n+1)$ is a product of an odd and an even integer, so it is always even.

Since a_n is even for all $n \geq 1$, it is trivially eventually even, which also means it is frequently even, so (1) and (3) are true.

Additionally, (2) and (4) are false because a_n being frequently odd would require there to be infinitely many odd terms in (a_n) (which there are obviously not), and a_n not being frequently odd implies that it is not eventually odd, either. $\square$

Problem 3. Let a_n be some sequence. Say if the following statements are true or false, and justify your answer.

- (1) If $a_n \rightarrow +\infty$ then a_n must be eventually increasing.
- (2) If $a_n > 0$ and a_n is unbounded then a_n must diverge to $+\infty$.
- (3) If $a_n \rightarrow 0$ then either $a_n \rightarrow 0^+$ or $a_n \rightarrow 0^-$.
- (4) If $a_n \rightarrow 0^+$ then a_n must be eventually decreasing.

Solution. (1) is false: consider the sequence (a_n) given by $(2, 1, 4, 3, 6, 5, 8, 7, \dots)$, which alternates between the smallest positive even number not currently in the sequence and the smallest positive odd number not currently in the sequence. This is clearly unbounded, since for any M , $n > 100! \cdot |M| + 10^{10}$ implies $a_n > M$ (in particular $a_n > n - 1$ for all n , so the bound on n is slightly overkill).

(2) is false: take $b_n = 2^n$ if n is even and 1 if n is odd. Then, all terms of the sequence are positive and, since $\lim_{k \rightarrow \infty} 2^k = +\infty$, b_n must be unbounded (otherwise the powers of two would have an upper bound). However, this only means that, for each real M , b_n is *frequently* greater than M ; for the sequence to diverge to $+\infty$, it would need to be *eventually* greater than M , which does not hold if, say, $M = \pi$ (any $M \geq 1$ works as a counterexample).

(3) is false: take $c_n = \frac{(-1)^n}{n}$. This alternates between being positive and negative, but its limit is 0, as shown in textbook problem 8.1(a).

(4) is false: take $d_n = \frac{1}{a_n}$ (where (a_n) is the sequence $(2, 1, 4, 3, 6, 5, \dots)$ defined earlier). For each ε , $|d_n - 0| = c_n = \frac{1}{a_n} < \varepsilon$ is eventually true because it holds whenever $n > \frac{100!}{\varepsilon} + 1000$, since $a_n > n - 1$ implies

$$d_n < \frac{1}{n-1} = \frac{1}{\frac{100!}{\varepsilon} + 999} < \frac{\varepsilon}{100!} < \varepsilon.$$

This means that $c_n \rightarrow 0^+$, but the sequence is not eventually decreasing because $\frac{1}{2k} < \frac{1}{2k-1}$ for all positive integers k . $\square$